

Linear and Nonlinear Kalman Filtering: theory and applications

SCUDO Ph.D. course A.Y. 2026

Marco Todescato

Fraunhofer Italia Research SCARL

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1 General info about the course and Introduction

Schedule

Date	Time	Tentative Topic
27/04/2026	h10.30-12:30 (2h)	Intro e Preliminaries
04/05/2026	h09.30-12:30 (3h)	Preliminaries/Linear Theory
11/05/2026	h09.30-12:30 (3h)	Linear Theory
18/05/2026	h09.30-13:30 (4h)	Linear Theory + Essay
25/05/2026	h09.30-13:30 (4h)	Linear/Nonlinear Theory + Essay
01/06/2026	h09.30-13:30 (4h)	Nonlinear Theory/Adds-on + Essay

Practical coding essay + report:

- on a selected application: pick a topic possibly related to your need/research.
- partially developed during class.
- programming language: python – either using jupyter notebooks (recommended) or script based. Make sure all necessary dependencies to run are available.

For other programming language ask first!!

- no use of predefined libraries with KF implementations.
- you can work in a group if preferred.
- **objective**: explore (possibly all) the topics covered during the course, from linear to non-linear KF. You might need to start from modeling assumptions, linearize if necessary, discuss noise statistics and then implement the filter. Try to explore limits and nuances as if it were a small research project.
- present results in a pdf report.

- IT Picci, G. (2007). Filtraggio statistico (Wiener, Levinson, Kalman) e applicazioni. Libreria Progetto.**
- EN Sanso F., Pileggi, C. and Biagi, L. (2024). A tutorial on Kalman filtering. The linear, extended and unscented filters. (pdf available as resource in the material folder)**
- EN Anderson, B. D., and Moore, J. B. (2005). Optimal filtering. Courier Corporation.**

G-drive: https://drive.google.com/drive/folders/1BbuSTi04wym0q0AqeLUkiVum_14kRRzD?usp=drive_link

github: https://github.com/MarcoTodescato/linear_and_nonlinear_kalman_filtering

Goal

How to formulate estimation problems encountered in many engineering fields and applications as instances of a general dynamical system state estimation problem and how to derive the mathematical solution of the latter problem (possibly depending on additional assumption on the functional family of the considered dynamical system).

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Essentially, this will lead to the well-known (linear) Kalman filter and its (nonlinear) extensions.

A bit of history: the theories of statistical filtering

Wiener-Kolmogorov (1940)

- $I = (-\infty, t]$, infinite time support
- $\{\mathbf{x}(t)\}$, $\{\mathbf{y}(t)\}$ are weakly jointly stationary

The impulse response of the optimal filter is time-invariant hence it can be analytically computed once and for all times, knowing the processes statistics.

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Levinson (1950)

- no more infinite (lower unbounded) time support

The impulse response of the filter is time-varying but can be efficiently computed at time t using a recursive structure which exploits the computations until $t - 1$.

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Kalman (1960)

- signals can be modeled as stochastic dynamical systems of finite size

Looks for recursive finite-memory schemes that resembles that same dynamical system structure of finite size.

A bit of history: a change of perspective

Revolutionary approach introduced by R.E. Kalman in 1960 to the problem of statistical filtering.

Change of perspective:

- **Wiener-Kolmogorov**: explicit computation of the filter **transfer function**, practically useful only for signals characterized by a **rational** spectral density
 - implicit computations possible only for rational signals
 - actual physical implementation of the filter as a set of difference (recursive) equations only for rational representations

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 - actual physical implementation of the filter as a set of difference (recursive) equations only for rational representations
- **Kalman**: hypothesis that the signals can be described as (linear) **dynamical systems** of finite dimension
 - no transfer functions involved
 - natural formulation of the filter as a fixed algorithmic structure essentially consisting of a discrete-time (linear) dynamical system of finite memory, driven by measurements
 - implementation always possible as a recursive algorithm

Prediction vs Filtering vs Smoothing

Assume $\{\mathbf{x}(t)\}$ and $\{\mathbf{y}(t)\}$ are (second order¹) processes of known dimensions.
Denote with $\hat{\mathbf{x}}(t|I)$ the estimator of $\mathbf{x}(t)$ given $\{\mathbf{y}(t)\}_{t \in I}$, I being the time support.
It is possible to identify 3 classes of estimation problems.

¹First and second moment statistics exist and are finite.

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Prediction

Find $\hat{\mathbf{x}}(t + h|I)$ being $I = [t_0, t]$ and $h > 0$, that is the estimator h -step ahead in the future from measurements up to time t .

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Smoothing

Find $\hat{\mathbf{x}}(t|I)$ being $I = [t_0, t_1]$ with $t < t_1$, that is the estimator at time t from all available measurements up to t_1 in the future.

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- 2 Preliminaries: before diving in... some definitions and preliminary results

Random signal

A random signal can be defined as the tuple (T, V, S) where

- T is the set of time. We consider discrete-time signals, $T = \mathbb{Z}$
- V is the “alphabet” of signal values, $V = \mathbb{R}$ or $V = \mathbb{R}^n$
- S is the sample space. Every $s \in S$ is a function $s : t \mapsto s(t)$ where $t \in T$ and $s(t) \in V$ also called *trajectory* of the signal

Random signals and stochastic processes

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Stochastic process

Mathematical formulation/description used to formally model the aleatoric nature of a random signal. Consisting of the tuple $(S, \mathcal{S}, P)$ where:

- S sample space. Set of all possible trajectories
- $\mathcal{S}$ events set of all logical propositions or events regarding the signal
- P probability measure on $\mathcal{S}$, i.e. $P : \mathcal{S} \mapsto [0, 1]$ mapping every $A \in \mathcal{S}$ to $P(A) \in [0, 1]$, the a-priori “confidence” A actually happening

A more general formulation

Stochastic process (usual def in prob. theory)

A collection of random variables, $\{y(t)\}$, indexed by a temporal index $t \in T$, all defined on the same (abstract) probability space $(\Omega, \mathcal{A}, P)$ where:

- Ω event set of “state of nature” ω
- $\mathcal{A}$ σ -algebra of subsets of Ω
- P probability measure on $\mathcal{A}$

Each variable is a ($\mathcal{A}$ -measurable) function $y(t) : t \mapsto y(t, \omega)$. The *trajectories* are the successive determinations $\{y(\cdot, \tilde{\omega})\}$ of $y(t)$ corresponding to a particular event $\tilde{\omega} \in \Omega$

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Our definition $(S, \mathcal{S}, P)$ is a “concrete” version of the abstract $(\Omega, \mathcal{A}, P)$. If we define a variable $y(t) \in (S, \mathcal{S}, P)$ as

$$y(t, s) := s(t)$$

where $y(t)$ associates at every trajectory $s \in S$ its value at t , then it is clear that, for varying t , the successive determinations of the variables $\{y(t)\}$ corresponding to the event $\bar{s} \in \mathcal{S}$ describe indeed the trajectory $\{\bar{s}(t)\}$ of the process.

Let's clarify...

The probabilistic formulation and formalization introduced is *just* a convenient choice to describe (or model) the unpredictable nature of a signal. It is worth remembering that this is one possible choice that eventually allows one to resort to tools from probability theory to describe, handle and develop useful tools for filtering and identification. For this reason, it is important to understand that, regardless the real nature of the underline signal, the same formulation (as *stochastic process*) can indeed be used to describe every signal (aleatoric or not). On the other hand, it might be worth remembering that this choice is dictated by the “simplicity”, compactness and generality to describe events that could indeed also be described using a “deterministic” formulation.

The Problem of *Estimation*: Estimate vs Measure

Statistical problem of estimation seen as formalization of the mathematical problem of measuring where the goal is that of getting information about a desired quantity x , non directly observable, based on observable data, y , whose values are produced by the measurement “instrument”.

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Ideally, knowing the “physical coupling”, i.e., relation, between x and y should allow to retrieve the value of x based on observable values of y . Practically, this relation is either not perfectly known or influenced by *environmental conditions* which introduce *uncertainty* in the measurement process.

$$y = h(x) \iff y = h(x) + \epsilon$$

As a consequence, the objective can be understood (or interpreted) as that of *estimating* the value of x based on the observed values of y .

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Following this “uncertainty-based” interpretation:

- it is not always straightforward how to generally get x from y
- being impractical (meaningless) to reconstruct x with certainty, how to *best* approximate it?

The Bayesian approach

x assumed to be a random quantity characterized by some a-priori information

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Notation: random variables vs sample values

It is important to understand the difference between a random variable, for which bold letters $\mathbf{x}$ will be used vs the corresponding sample values, i.e., the actual value taken by the random variable, for which normal case letters x will be used.

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Bayesian inference

Assume there exist a random variable/vector $\mathbf{x}$ described by the probability density $p_{\mathbf{x}}(\mathbf{x})$ and the random variable/vector $\mathbf{y}$ of measurements described by the conditional density $f(\mathbf{y}|\mathbf{x}) = f(\mathbf{y}|\mathbf{x} = \mathbf{x}) = \frac{P(\mathbf{y} \leq \mathbf{y} \leq \mathbf{y} + d\mathbf{y} | \mathbf{x} = \mathbf{x})}{d\mathbf{y}}$.

The problem of Bayesian inference/estimation is that of reconstructing $\mathbf{x}$ based on $p_{\mathbf{x}}$, $f(\mathbf{y}|\mathbf{x})$ and the observation of $\mathbf{y} = \mathbf{y}$. This can be understood as to:

- compute the new distribution of $\mathbf{x}$ determined by the observation $\mathbf{y} = \mathbf{y}$, or
- reconstruct $x = \mathbf{x}(\omega)$ determined by the “measurement” process

Bayesian inference: complete reconstruction

Assuming $p_{\mathbf{x}}(x)$ and $f(y|x)$ completely known, the problem is that of computing

$$f(x|y) = \frac{f(y|x)p_{\mathbf{x}}(x)}{\int f(y|x)p_{\mathbf{x}}(x)dx} = \frac{p_{y\mathbf{x}}(y, x)}{p_y(y)}$$

i.e., the conditional *a-posteriori* density of $\mathbf{x}$ given $\mathbf{y} = y$, which is the most exhaustive and complete description possible.

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Explicit analytical computation of the above equations often not possible.

In practice, this is rarely the case and/or even necessary because rather than an exhaustive description, a more handy point estimate of $\mathbf{x}$, telling us what is the unobservable sample value $\mathbf{x} = x$ that gave rise to the observation $\mathbf{y} = y$, is more useful.

Bayesian inference: point estimate

Assume the relation between $\mathbf{x}$ and $\mathbf{y}$ could analytically be described as

$$\mathbf{y} = h(\mathbf{x}, \mathbf{w})$$

being $\mathbf{w}$ the random vector capturing the variable interactions with the environment, i.e., the “measurement noise”.

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During the measurement process the particular experimental conditions ω give raise to a specific determination $\mathbf{w}(\omega) = \mathbf{w}$ which, in turn, give raise to an observed value $\mathbf{y} = y$ linked to $\mathbf{x} = x$ by the relation

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Then, the problem is that of reconstructing the value $\mathbf{x}$ of $\mathbf{x}$ that “caused” $\mathbf{y} = \mathbf{y}$ during the measurement process. Since estimating/knowing $\mathbf{w}$ is intrinsically impossible, the problem is understood as that of “best” approximating $\mathbf{x}$.

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It is then necessary to define a “reasonable” class of approximation criteria.

Cost/Loss Function

It is called “loss/cost” a scalar function $c : \mathbb{R}^n \mapsto \mathbb{R}$ of $\xi = z - x$, $z \in \mathbb{R}^n$, s.t. $c \geq 0$, $c'' > 0$ (strictly convex) and $c(0) = 0$. It is said symmetric if $c(-\xi) = c(\xi)$.

E.g.: $c(z - x) = \|z - x\|_Q^2$ where $Q > 0$ symmetric and $\|x\|_Q^2 = x^T Q x$.

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z plays the role of approximation of x . The problem could then be that of finding z that minimizes c for every value of the unknown x . In practice it is “smarter” to consider all the available probabilistic information, e.g., given by $\mathbf{y} = y$.

Bayesian inference: conditional/posterior expected risk

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Conditional Expected Risk and Point Bayesian Estimate

Consider the *Conditional Expected Risk*

$$R(z, y) := \mathbb{E}[c(z - \mathbf{x}) | \mathbf{y} = y] = \int_{\mathbb{R}^n} c(z - x) f(x|y) dx$$

Then, the *Point Bayesian Estimate* is defined as

$$\hat{x} = \text{Arg min}_z R(z, y)$$

Theorem: Conditional Expectation

If the loss c is symmetric and the conditional density $f(\cdot|y)$ is symmetric around its mean $\mu(y) = \mathbb{E}[\mathbf{x}|\mathbf{y} = y] = \int \mathbf{x}f(\mathbf{x}|y)d\mathbf{x}$, then $\hat{\mathbf{x}}$ does not depend on c and it is equal to the conditional mean of $\mathbf{x}$ given $\mathbf{y} = y$, i.e., $\mathbb{E}[\mathbf{x}|\mathbf{y} = y]$.

Moreover, if the cost c is of quadratic form, i.e., $\|\cdot\|_Q^2$, then $\hat{\mathbf{x}} = \mathbb{E}[\mathbf{x}|\mathbf{y} = y]$ for all types of conditional density $f(\mathbf{x}|\mathbf{y})$ (assuming the conditional mean exists).

More generally, among all possible measurable functions $g : \mathbb{R}^m \mapsto \mathbb{R}^n$, for every symmetric $Q > 0 \in \mathbb{R}^{n \times n}$,

$$\mathbb{E}[\mathbf{x}|\mathbf{y}] = \text{Arg min}_{g(\cdot)} \mathbb{E}[\|\mathbf{x} - g(\mathbf{y})\|_Q^2]$$

²Estimator: function (algorithm) of the observed data mapping them to point estimates. 

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Interpreted as a function of y , for symmetric c and f , the conditional mean $\mathbb{E}[\mathbf{x}|\mathbf{y}] : y \mapsto \mathbb{E}[\mathbf{x}|\mathbf{y} = y]$ is the Bayesian estimator² $\hat{\mathbf{x}}$ of $\mathbf{x}$ that minimizes the conditional expected risk $R(z, y)$. This is also true for any f if c is quadratic. In this case it is referred to as the *minimum squared Bayesian estimator*.

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Theorem: Conditional Expectation (proof of last part)

For the law of total expectation, we can write

$$\begin{aligned}\mathbb{E}[\|\mathbf{x} - g(\mathbf{y})\|_Q^2] &= \int_{\mathbb{R}} p_{\mathbf{y}}(y) \mathbb{E}[\|\mathbf{x} - g(\mathbf{y})\|_Q^2 | \mathbf{y} = y] dy \\ &= \mathbb{E} [\mathbb{E}[\|\mathbf{x} - g(\mathbf{y})\|_Q^2 | \mathbf{y}]] \\ &= \mathbb{E} [\mathbb{E}[\|\mathbf{x} - \mathbb{E}[\mathbf{x} | \mathbf{y}] + \mathbb{E}[\mathbf{x} | \mathbf{y}] - g(\mathbf{y})\|_Q^2 | \mathbf{y}]] \\ &= \mathbb{E} \left[\mathbb{E}[(\mathbf{x} - \mathbb{E}[\mathbf{x} | \mathbf{y}])^T Q (\mathbf{x} - \mathbb{E}[\mathbf{x} | \mathbf{y}]) | \mathbf{y}] \right. \\ &\quad \left. + 2\mathbb{E}[(\mathbf{x} - \mathbb{E}[\mathbf{x} | \mathbf{y}])^T Q (\mathbb{E}[\mathbf{x} | \mathbf{y}] - g(\mathbf{y})) | \mathbf{y}] \right. \\ &\quad \left. + \mathbb{E}[(\mathbb{E}[\mathbf{x} | \mathbf{y}] - g(\mathbf{y}))^T Q (\mathbb{E}[\mathbf{x} | \mathbf{y}] - g(\mathbf{y})) | \mathbf{y}] \right] \\ &= \mathbb{E}[\|\mathbf{x} - \mathbb{E}[\mathbf{x} | \mathbf{y}]\|_Q^2] + \mathbb{E}[\|\mathbb{E}[\mathbf{x} | \mathbf{y}] - g(\mathbf{y})\|_Q^2]\end{aligned}$$

where the last equality holds by taking the mean and since the second double product term is zero conditioning on $\mathbf{y}$. And since the two terms are non-negative and the first does not depend on g , we can minimize the second by setting $g(\mathbf{y}) = \mathbb{E}[\mathbf{x} | \mathbf{y}]$.

Bayesian inference: MAP estimate

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What if, instead of minimizing risk, one wants to maximize gain?

Expected gain

$$G(z, y) := \mathbb{E}[(z - \mathbf{x}) | \mathbf{y} = y] = \int_{\mathbb{R}^n} \gamma(z - x) f(x|y) dx$$

being γ a concave function with its max in the origin.

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By choosing γ to arbitrarily approximate Dirac's impulse $\delta \rightarrow G(z, y) \approx f(z|y)$

The corresponding Max Expected Gain (or *Maximum a Posterior* - M.A.P.) estimator is given by

$$\hat{x}_{MAP}(y) := \text{Arg max}_z f(z|y)$$

which, for general $f(x|y)$ is the **conditional mode** ($\equiv \mathbb{E}[\mathbf{x}|\mathbf{y}]$ for unimodal symmetric densities).

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The corresponding Max Expected Gain (or *Maximum a Posterior* - M.A.P.) estimator is given by

$$\hat{x}_{MAP}(y) := \text{Arg max}_z f(z|y)$$

which, for general $f(x|y)$ is the **conditional mode** ($\equiv \mathbb{E}[\mathbf{x}|\mathbf{y}]$ for unimodal symmetric densities).

Unfortunately analytical computations of mean (or mode) are usually not possible.

Bayesian inference: the Gaussian case

The Gaussian case

If $\mathbf{x}$ and $\mathbf{y}$ are jointly Gaussian described by a density function with mean and variance

$$\mu_z = \begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}; \quad \Sigma_z = \begin{bmatrix} \Sigma_x & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_y \end{bmatrix}$$

the conditional density is still Gaussian with mean and variance (if $\Sigma_y > 0^a$)

$$\begin{aligned} \mathbb{E}[\mathbf{x}|\mathbf{y}] &= \mu_x + \Sigma_{xy}\Sigma_y^{-1}(\mathbf{y} - \mu_y) \\ \text{Var}[\mathbf{x}|\mathbf{y}] &= \Sigma_x - \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx} \end{aligned}$$

^aIf this is not the case it can be shown that using the pseudo-inverse $(\cdot)^\dagger$ works.

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Interpreting $\mathbb{E}[\mathbf{x}|\mathbf{y}]$ as the Bayesian estimator of $\mathbf{x}$ given $\mathbf{y}$, notice that:

- the estimation error is defined as: $\tilde{\mathbf{x}} := \mathbf{x} - \mathbb{E}[\mathbf{x}|\mathbf{y}]$
- $\tilde{\mathbf{x}}$ is independent from the data, i.e., all available info has been exploited
- $\text{Var}[\mathbf{x}|\mathbf{y}]$ (cond. var. of $\mathbf{x}|\mathbf{y}$) can be understood as $\Sigma_{\tilde{\mathbf{x}}}$ (uncond. var. of $\tilde{\mathbf{x}}$), does not depend on the data, and is given by the difference between Σ_x and $\text{Var}\mathbb{E}[\mathbf{x}|\mathbf{y}] = \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx}$ (uncond. vars. of $\mathbf{x}$, $\mathbb{E}[\mathbf{x}|\mathbf{y}]$, resp.)

The Gaussian case: proof

Consider $\bar{\mathbf{z}} := \mathbf{z} - \mu_{\mathbf{z}} \sim \mathcal{N}(0, \Sigma_{\mathbf{z}})$. Consider a linear transformation of $\tilde{\mathbf{x}} = \bar{\mathbf{x}} + A\bar{\mathbf{y}}$, $\tilde{\mathbf{y}} = \bar{\mathbf{y}}$ such that $\tilde{\mathbf{x}} \perp \tilde{\mathbf{y}}$. That is

$$\mathbb{E}[\tilde{\mathbf{x}}\tilde{\mathbf{y}}^T] = 0 = \Sigma_{\mathbf{xy}} + A\Sigma_{\mathbf{y}} \implies A = -\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{y}}^{-1}$$

Clearly, since linear transforms maintain Gaussianity,

$$\tilde{\mathbf{z}} \sim \mathcal{N}(0, \Sigma_{\tilde{\mathbf{z}}}) = \begin{bmatrix} I & A \\ & I \end{bmatrix} \Sigma_{\mathbf{z}} \begin{bmatrix} I & \\ A^T & I \end{bmatrix} = \begin{bmatrix} \Sigma_{\tilde{\mathbf{x}}} & \\ & \Sigma_{\mathbf{y}} \end{bmatrix}, \quad \Sigma_{\tilde{\mathbf{x}}} = \Sigma_{\mathbf{x}} - \Sigma_{\mathbf{xy}}\Sigma_{\mathbf{y}}^{-1}\Sigma_{\mathbf{yx}}$$

and being Gaussian and uncorrelated, $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{y}}$ are also independent

$$\begin{aligned} \implies \bar{\mathbf{x}} = \tilde{\mathbf{x}} - A\bar{\mathbf{y}} &\implies \mathbb{E}[\bar{\mathbf{x}}|\bar{\mathbf{y}}] = -A\bar{\mathbf{y}} = \Sigma_{\mathbf{xy}}\Sigma_{\mathbf{y}}^{-1}\bar{\mathbf{y}} \\ \text{Var}[\bar{\mathbf{x}}|\bar{\mathbf{y}}] &= \Sigma_{\tilde{\mathbf{x}}} = \Sigma_{\mathbf{x}} - \Sigma_{\mathbf{xy}}\Sigma_{\mathbf{y}}^{-1}\Sigma_{\mathbf{yx}} \end{aligned}$$

Introducing the means you get the final result.

Conditional Expectation: brief recap

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- $\mathbb{E}[\mathbf{x}|\mathbf{y}]$ is the function $g(\cdot)$ of the data $\mathbf{y}$ that minimizes $\mathbb{E}\|\mathbf{x} - g(\mathbf{y})\|_Q^2$ among the (vast) class of measurable functions that are also unbiased, i.e., $\mathbb{E}[g(\mathbf{y})] = \mathbb{E}[\mathbf{x}]$;

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- for $Q = I$, $\mathbb{E}\|\mathbf{x} - g(\mathbf{y})\|_Q^2$ is the scalar variance of the estimation error defined as $\tilde{\mathbf{x}} := \mathbf{x} - \mathbb{E}[\mathbf{x}|\mathbf{y}] \implies \mathbb{E}[\mathbf{x}|\mathbf{y}]$ is the *minimum variance estimator* of $\mathbf{x}$;

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What if the functional family of $g(\cdot)$ is a-priori restricted? E.g. to be linear

Linear Minimum Mean Squared Error (MMSE) Estimators

Restricting $g(\cdot)$ to the family of linear unbiased functions of $\mathbf{y}$, we have

Linear MMSE Estimator: the Problem (vector form)

The Linear MMSE Estimator is the linear function of the data

$$g(\mathbf{y}) = A\mathbf{y} + b, \quad A \in \mathbb{R}^{n \times m}, \quad b \in \mathbb{R}^n$$

that minimizes the mean of the squared estimation error $\mathbb{E}\|\mathbf{x} - g(\mathbf{y})\|^2$, that is

$$\min_{A,b} \mathbb{E}\|A\mathbf{y} + b - \mathbf{x}\|^2.$$

Minimum introduction to Hilbert spaces

A Hilbert space $\mathbf{H}$ is a vector space (over $\mathbb{R}$ or $\mathbb{C}$) equipped with an inner product $\langle \cdot, \cdot \rangle$ which is complete w.r.t. the norm $\|x\| = \sqrt{\langle x, x \rangle}$ induced by the inner product. That is, every Cauchy sequence converges with respect to $\|\cdot\|$.

Examples:

- $\mathbb{R}^n$ endowed with the standard Euclidean inner product;
- $L^2(\Omega, \mathcal{A}, P)$ of second order variables of a random experiment $\{\Omega, \mathcal{A}, P\}$: the space of variables f , $\mathbb{E}[|f|^2] < \infty$ (mean squared integrable $\equiv$ finite variance), with inner product $\langle \xi, \eta \rangle = \mathbb{E}[\xi \bar{\eta}]$ given by the correlation between variables;
- $\mathbf{H}(\mathbf{y}) := \text{span} \{\mathbf{y}_1, \dots, \mathbf{y}_m\} = \left\{ \sum_{i=1}^m a_i \mathbf{y}_i; a_i \in \mathbb{R} \right\} \subseteq L^2(\Omega, \mathcal{A}, P)$ linearly generated by the second order random process $\mathbf{y}$ (collection of variables) with $\mathbb{E}[\mathbf{y}_i] = 0$ (zero mean) and $\|\mathbf{y}_i\|_{\mathbf{H}}^2 = \mathbb{E}[|\mathbf{y}_i|^2] = \text{Var}[\mathbf{y}_i] < \infty$ (finite variance).

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$\Rightarrow$ resorting to Hilbert spaces and, in particular, $\mathbf{H}(\mathbf{y})$ allows to interpret the problem geometrically!

Theorem: Orthogonal Projections (scalar form)

Let $\mathbf{x} \in \mathbf{H}$ be a (scalar) random variable. Let $\mathbf{H}(\mathbf{y}) \subset \mathbf{H}$. The variable $\mathbf{z}^* \in \mathbf{H}(\mathbf{y})$ minimizing $\|\mathbf{x} - \mathbf{z}\|_{\mathbf{H}}^2 = \mathbb{E}[|\mathbf{x} - \mathbf{z}|^2]$ is unique and is the orthogonal projection of $\mathbf{x}$ onto $\mathbf{H}(\mathbf{y})$. Necessary and sufficient condition for $\mathbf{z}^*$ to be the orthogonal projection is that

$$\mathbf{x} - \mathbf{z}^* \perp \mathbf{H}(\mathbf{y}) \quad \equiv \quad \mathbb{E}[(\mathbf{x} - \mathbf{z}^*)\mathbf{y}_i] = 0 \quad i \in \{1, \dots, m\}$$

being $\mathbf{y}_i$ the generators of $\mathbf{H}(\mathbf{y})$.

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Thanks to this fundamental theorem, the properties inherited from Hilbert spaces and the correspondence between the notion of orthogonality ($\perp$) and uncorrelation (through the inner product/norm) a solution to the *Linear MMSE Estimator* can be drawn (directly in vector form from a element-wise application of the projection theorem).

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Linear MMSE Estimator: the Solution (vector form)

It can be shown that the solution $\hat{\mathbf{x}}(\mathbf{y}) = g^*(\mathbf{y})$ (also denoted $\hat{\mathbb{E}}[\mathbf{x}|\mathbf{y}]$) is given by

$$\hat{\mathbf{x}}(\mathbf{y}) = \mu_{\mathbf{x}} + \Sigma_{\mathbf{xy}}\Sigma_{\mathbf{y}}^{-1}(\mathbf{y} - \mu_{\mathbf{y}})$$

that is $A^* = \Sigma_{\mathbf{xy}}\Sigma_{\mathbf{y}}^{-1}$, $b^* = \mu_{\mathbf{x}} - A^*\mu_{\mathbf{y}}$. Equivalently to the Gaussian case, it depends only of the first and second moments of the joint density.

$$\hat{\mathbb{E}}[\mathbf{x}|\mathbf{y}] \neq \mathbb{E}[\mathbf{x}|\mathbf{y}]$$

- $\mathbb{E}[\mathbf{x}|\mathbf{y}]$ minimum variance estimator = conditional mean
- $\hat{\mathbb{E}}[\mathbf{x}|\mathbf{y}]$ linear minimum variance estimator $\neq$ conditional mean
- $\hat{\mathbb{E}}[\mathbf{x}|\mathbf{y}]$ in general depends only on up to second order statistics
- $\hat{\mathbb{E}}[\mathbf{x}|\mathbf{y}]$ sometimes referred to as **weak conditional mean**
- $\hat{\mathbb{E}}[\mathbf{x}|\mathbf{y}] = \mathbb{E}[\mathbf{x}|\mathbf{y}]$ for jointly Gaussian densities
- $\hat{\mathbb{E}}[\mathbf{x}|\mathbf{y}]$ very “far” from conditional mean for general densities

Example with code

Before looking at more theory ... a first example of Kalman Filtering!

Jupyter notebook: `linear_constant_velocity.ipynb`

2 Linear Theory: Minimum Variance Linear Estimator for Stochastic Processes

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Considerations

1. Linear Estimators $\implies$ depending on first and second statistics, sufficient to find estimators minimizing squared measures of the estimation errors, e.g., variance.
2. Processes $\implies$ so far we considered $\mathbf{x}$ and $\mathbf{y}$ of fixed known dimensions. We now turn to the more interesting (and more useful) setting of “continuous” flow of new measurements $\{\mathbf{y}(t)\}$ of a (most often) time-varying signal $\{\mathbf{x}(t)\}$. In this scenario, the time interval I , during which data/measurements $\{\mathbf{y}(t)\}_{t \in I}$ flow, increases. Hence, it appears clearly how a suitable algorithmic structure to process an increasing amount of data becomes indeed fundamentally important.

Some more notation related to Hilber spaces...

- When dealing with (random) processes $\mathbf{y}(t)$ (collection of variables) over time, i.e., $\{\mathbf{y}_k(t), t \in \mathbb{Z}, k = \{1, \dots, m\}\}$, the Hilbert space $\mathbf{H}(\mathbf{y}) \subset \mathbf{H}$ generated by the (second order) process can be defined as

$$\begin{aligned}\tilde{\mathbf{H}}(\mathbf{y}) &= \left\{ \sum a_{ki} \mathbf{y}_k(t_i); a_{ki} \in \mathbb{R}, t_i \in \mathbb{Z}, k = \{1, \dots, m\} \right\} \\ \mathbf{H}(\mathbf{y}) &= \overline{\text{span}} \{ \mathbf{y}(t), t \in \mathbb{Z} \} = \text{cls} \{ \tilde{\mathbf{H}}(\mathbf{y}) \}\end{aligned}$$

with $\text{cls} \{ \cdot \}$ denoting the formal operation of closure that is completing the corresponding space with the limits of all its Cauchy sequences which, in the case of infinite dimensional space, could not own to the original space.³

³We will take these technical aspects for granted.

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- Considering past and present time $s \in (-\infty, t]$, we denote

$$\mathbf{H}_t^-(\mathbf{y}) := \overline{\text{span}} \{ \mathbf{y}(s); s \leq t \}; \quad \mathbf{H}_s^-(\mathbf{y}) \subseteq \mathbf{H}_t^-(\mathbf{y}), s \leq t; \quad \lim_{t \rightarrow \infty} \mathbf{H}_t^-(\mathbf{y}) = \mathbf{H}(\mathbf{y})$$

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- Considering present and future time $s \in [t, +\infty)$, we denote

$$\mathbf{H}_t^+(\mathbf{y}) := \overline{\text{span}} \{ \mathbf{y}(s); s \geq t \}; \quad \mathbf{H}_s^+(\mathbf{y}) \supseteq \mathbf{H}_t^+(\mathbf{y}), s \leq t; \quad \lim_{t \rightarrow -\infty} \mathbf{H}_t^+(\mathbf{y}) = \mathbf{H}(\mathbf{y})$$

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...and orthogonal projections

Given $\mathbf{H}(\mathbf{y}) \subset \mathbf{H}$, $\forall \mathbf{x} \in \mathbf{H}$

$$\mathbf{x} = \hat{\mathbf{x}} + \tilde{\mathbf{x}}$$

where $\hat{\mathbf{x}} \in \mathbf{H}(\mathbf{y})$ is the orthogonal projection (also denoted as $\hat{\mathbb{E}}[\mathbf{x}|\mathbf{H}(\mathbf{y})]$) of $\mathbf{x}$ onto $\mathbf{H}(\mathbf{y})$ and $\tilde{\mathbf{x}} \perp \mathbf{H}(\mathbf{y})$

- (idempotent) $\hat{\mathbb{E}} \left[\hat{\mathbb{E}}[\cdot|\mathbf{H}(\mathbf{y})] \right] = \hat{\mathbb{E}}[\cdot|\mathbf{H}(\mathbf{y})]$
- if $\mathbf{H}(\mathbf{y}_1) \subset \mathbf{H}(\mathbf{y}_2)$, $\hat{\mathbb{E}} \left[\hat{\mathbb{E}}[\cdot|\mathbf{H}(\mathbf{y}_2)]|\mathbf{H}(\mathbf{y}_1) \right] = \hat{\mathbb{E}}[\cdot|\mathbf{H}(\mathbf{y}_1)]$

In particular if $s < t$

$$\hat{\mathbb{E}} \left[\hat{\mathbb{E}}[\mathbf{x}|\mathbf{H}_t^-(\mathbf{y})]|\mathbf{H}_s^-(\mathbf{y}) \right] = \hat{\mathbb{E}}[\mathbf{x}|\mathbf{H}_s^-(\mathbf{y})]$$

State Space models

Let $\{\mathbf{y}(t)\}$ an m -dimensional stochastic process defined on $[t_0, \infty)$. We say it is a *finite dimensional process* if we can express it as

$$\begin{aligned}\mathbf{x}(t+1) &= A\mathbf{x}(t) + B\mathbf{n}(t) \\ \mathbf{y}(t) &= C\mathbf{x}(t) + D\mathbf{n}(t), \quad t \geq t_0, \quad \mathbf{x}(t_0) = \mathbf{x}_0\end{aligned}$$

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- $\{\mathbf{n}(t)\}$ is a (orthonormal) *white noise* process with zero mean and unit variance ($\mathbb{E}[\mathbf{n}(t)] = 0$, $\mathbb{E}[\mathbf{n}(t)\mathbf{n}(s)^T] = I\delta_{ts}$)

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- $\{\mathbf{x}(t)\}$ is a n -dimensional process, called *state process*, with initial value $\mathbf{x}_0$ which is uncorrelated w.r.t. all present and future values of $\mathbf{n}$, i.e., $\mathbb{E}[\mathbf{x}_0\mathbf{n}(t)^T] = 0$, $\forall t \geq t_0$ and $\mathbb{E}[\mathbf{x}_0] = \mu_0$, $\text{Var}[\mathbf{x}_0] = \Sigma_0$

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- A, B, C, D are *deterministic* matrices, possibly time varying but known⁴.
- $\{\mathbf{n}(t)\}$ is a (orthonormal) *white noise* process with zero mean and unit variance ($\mathbb{E}[\mathbf{n}(t)] = 0$, $\mathbb{E}[\mathbf{n}(t)\mathbf{n}(s)^T] = I\delta_{ts}$)
- $\{\mathbf{x}(t)\}$ is a n -dimensional process, called *state process*, with initial value $\mathbf{x}_0$ which is uncorrelated w.r.t. all present and future values of $\mathbf{n}$, i.e., $\mathbb{E}[\mathbf{x}_0\mathbf{n}(t)^T] = 0$, $\forall t \geq t_0$ and $\mathbb{E}[\mathbf{x}_0] = \mu_0$, $\text{Var}[\mathbf{x}_0] = \Sigma_0$

Essentially a finite dimensional process can be described as the output of a (time-varying) linear dynamical system driven by white noise.

⁴To avoid notation clutter, time dependence of the model matrices is omitted.

State Space models - the state process

Markov property

$\{\mathbf{x}(t)\}$ is a weak (sometimes also referred to as second-order) Markov process, i.e.,

$$\hat{\mathbb{E}}[\mathbf{x}(t)|\mathbf{H}_s(\mathbf{x})] = \hat{\mathbb{E}}[\mathbf{x}(t)|\mathbf{x}(s)] \quad \forall t \geq s$$

being $\mathbf{H}_s(\mathbf{x}) = \mathbf{H}_s^-(\mathbf{x})$ the Hilbert space generated by $\{\mathbf{x}(t_0), \dots, \mathbf{x}(s)\}$. If $\{\mathbf{n}(t)\}$ and $\mathbf{x}_0$ are jointly Gaussian, $\{\mathbf{x}(t)\}$ is strongly Markov (i.e., replace $\hat{\mathbb{E}}$ by $\mathbb{E}$, equivalently the entire distribution).

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Sketch of proof:

$$\mathbf{x}(t) = A^{t-t_0}\mathbf{x}_0 + \sum_{k=t_0}^{t-1} A^{t-1-k} B \mathbf{n}(k) = A^{t-s}\mathbf{x}(s) + \sum_{k=s}^{t-1} A^{t-1-k} B \mathbf{n}(k), \quad t \geq s$$

$$\begin{aligned} &\Rightarrow \{\mathbf{x}(t)\} \text{ linear in } \mathbf{x}_0, \{\mathbf{n}(t_0), \dots, \mathbf{n}(t-1)\} \Rightarrow \mathbf{x}(t) \perp \mathbf{H}_t^+(\mathbf{n}) \Rightarrow \mathbf{H}_s^-(\mathbf{x}) \perp \mathbf{H}_s^+(\mathbf{n}) \\ &\Rightarrow \hat{\mathbb{E}}[\mathbf{x}(t)|\mathbf{H}_s^-(\mathbf{x})] = \hat{\mathbb{E}}[\mathbf{x}(t)|\mathbf{x}(s)] = A^{t-s}\mathbf{x}(s) \end{aligned}$$

For the Gaussian case, $\hat{\mathbb{E}} \equiv \mathbb{E}$. Also the entire conditional probability depends on the past only through the conditional mean hence the property holds “globally”.

State Space models - the state process

The same “memory” property holds wrt the output process $\{\mathbf{y}(t)\}$

$$\hat{\mathbb{E}}[\mathbf{x}(t)|\mathbf{H}_s^-(\mathbf{x}, \mathbf{y})] = \hat{\mathbb{E}}[\mathbf{x}(t)|\mathbf{x}(s)]$$

$$\hat{\mathbb{E}}[\mathbf{y}(t)|\mathbf{H}_s^-(\mathbf{x}, \mathbf{y})] = \hat{\mathbb{E}}[\mathbf{y}(t)|\mathbf{x}(s)], \quad t \geq s$$

being $\mathbf{H}_s^-(\mathbf{x}, \mathbf{y}) = \mathbf{H}_s^-(\mathbf{x}) \vee \mathbf{H}_{s-1}^-(\mathbf{y})$ vector sum of $\mathbf{H}_s^-(\mathbf{x})$ (past and present wrt to s) and $\mathbf{H}_{s-1}^-(\mathbf{y})$ (strict past wrt to s).

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Conditional Incorelation (Orthogonality)

Define $\mathbf{z}(t) := [\mathbf{x}(t+1)^T, \mathbf{y}(t)^T]^T$,

$$\mathbf{H}_t^-(\mathbf{z}) := \mathbf{H}_{t+1}^-(\mathbf{x}) \vee \mathbf{H}_t^-(\mathbf{y}), \quad \mathbf{H}_t^+(\mathbf{z}) := \mathbf{H}_{t+1}^+(\mathbf{x}) \vee \mathbf{H}_t^+(\mathbf{y})$$

$$\mathbf{X}_t := \text{span} \{\mathbf{x}_1(t), \dots, \mathbf{x}_n(t)\} \quad \text{state space of } \mathbf{x} \text{ at } t$$

then

$$\mathbf{H}_t^-(\mathbf{z}) \perp \mathbf{H}_t^+(\mathbf{z}) \mid \mathbf{X}_t \implies \hat{\mathbf{z}}(t) = \hat{\mathbb{E}}[\mathbf{z}(t)|\mathbf{x}(t)] = \begin{bmatrix} A \\ C \end{bmatrix} \mathbf{x}(t)$$

State Space models - moments

From $\mathbb{E}[\mathbf{n}] = 0$ and previous (conditional) incorrelation properties, the evolution of mean and covariance of a linear finite dimensional system is described by

$$\mu_{\mathbf{x}}(t+1) = A\mu_{\mathbf{x}}(t), \quad \mu_{\mathbf{x}}(t_0) = \mu_0$$

$$\mu_{\mathbf{y}}(t) = C\mu_{\mathbf{x}}(t)$$

$$\Sigma_{\mathbf{x}}(t, s) = \begin{cases} A^{t-s}\Sigma(s), & t \geq s \\ \Sigma(t)(A^T)^{s-t}, & t \leq s \end{cases}$$

$$\Sigma_{\mathbf{y}}(t, s) = \begin{cases} CA^{t-s-1}G(s), & t > s \\ C\Sigma(t)C^T + DD^T, & t = s \\ G(t)^T(A^T)^{s-t-1}C^T, & t < s \end{cases}$$

$$G(s) = A\Sigma(s)C^T + BD^T$$

and $\Sigma(t) = \text{Var}[\mathbf{x}(t)]$ is the solution of the discrete Lyapunov equation

$$\Sigma(t+1) = A\Sigma(t)A^T + BB^T, \quad \Sigma(t_0) = \Sigma_0$$

State Space models - stability and (asymptotic) stationarity

Stability and (asymptotic) stationarity⁵ of the underlying processes play an important role in the relation between the Wiener-Kolmogorov and Kalman theories.

⁵Evolution of covariance depends on $\tau = t - s$, $\Sigma(s, t) = \Sigma(\tau)$, for $t, s \geq t_0$ and $t - t_0 \rightarrow \infty$.

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Assuming A, B, C, D are constant, if $\rho(A) < 1$ (asymptotically stable):

- $\mu_{\mathbf{x}}(t+1) = A\mu_{\mathbf{x}}(t) \rightarrow 0$

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- If $\mathbf{x}_0 \sim (0, \Sigma_{\infty})$, then $\{\mathbf{x}(t)\}, \{\mathbf{y}(t)\}$ are weakly stationary ($\forall t \geq t_0$) with

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Under the above conditions^a the two approaches (Wiener-Kolmogorov and Kalman) are essentially equivalent.

^aFor finite LTI systems, stationarity translates into rationality of the transfer function and, in turn, ARMA/ARMAX representation. Hence the possibility to practically compute the filter transfer function.

⁵Evolution of covariance depends on $\tau = t - s$, $\Sigma(s, t) = \Sigma(\tau)$, for $t, s \geq t_0$ and $t - t_0 \rightarrow \infty$.

State Space models - a common alternative notation

$$\begin{aligned}\mathbf{x}(t+1) &= A\mathbf{x}(t) + B\mathbf{n}(t), & \mathbb{E}[\mathbf{n}(t)] &= 0, \mathbb{E}[\mathbf{n}(t)\mathbf{n}(s)^T] = I\delta_{ts} \\ \mathbf{y}(t) &= C\mathbf{x}(t) + D\mathbf{n}(t), & t \geq t_0, \mathbf{x}(t_0) &= \mathbf{x}_0\end{aligned}$$

State Space models - a common alternative notation

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By defining $\mathbf{v}(t) := B\mathbf{n}(t)$ (state/process noise) and $\mathbf{w}(t) := D\mathbf{n}(t)$ (measurement/observation noise) and $\mathbf{z}(t) = [\mathbf{x}(t)^T \quad \mathbf{y}(t-1)^T]^T$, we have

$$\mathbf{z}(t+1) = \begin{bmatrix} A \\ C \end{bmatrix} \mathbf{z}(t) + \begin{bmatrix} \mathbf{v}(t) \\ \mathbf{w}(t) \end{bmatrix}, \quad \mathbb{E} \begin{bmatrix} \mathbf{v}(t) \\ \mathbf{w}(t) \end{bmatrix} = 0, \quad \text{Var} \begin{bmatrix} \mathbf{v}(t) \\ \mathbf{w}(t) \end{bmatrix} = \begin{bmatrix} Q & S \\ S^T & R \end{bmatrix}$$

Often, in practical scenarios, $S = 0$ i.e. uncorrelated process and measurement noise.

(Some) Realization techniques

(With a change of notation) Consider a discrete time ARMAX (Auto-Regressive Moving Average Exhogenous) model, extremely common to represent input/output relations:

$$A(z^{-1}) y_k = B(z^{-1}) u_k + C(z^{-1}) e_k, \quad e_k \sim (0, \sigma^2)$$

$$A(z^{-1}) = 1 + a_1 z^{-1} + \dots + a_{n_a} z^{-n_a},$$

$$B(z^{-1}) = b_0 + b_1 z^{-1} + \dots + b_{n_b} z^{-n_b},$$

$$C(z^{-1}) = 1 + c_1 z^{-1} + \dots + c_{n_c} z^{-n_c}.$$

- ARMA: $B = 0$ or $u = 0$
- ARX: $C = 1$

Assumptions (for simplicity):

- Scalar (input and output, SISO) processes. For MIMO processes, generalization using matrix polynomials
- $A(z^{-1})$ and $C(z^{-1})$ are already monic (higher order coeff. =1)

Realization: Controllable Canonical Form

The ARMAX description in transfer functions terms can be written:

$$y_k = \frac{B(z^{-1})}{A(z^{-1})} u_k + \frac{C(z^{-1})}{A(z^{-1})} e_k$$

An equivalent state space form with state $x_k \in \mathbb{R}^{n_a}$ is

$$x_{k+1} = A_c x_k + B_c u_k + B_c e_k, \quad y_k = (C_u + C_e) x_k + D_u u_k + D_e e_k.$$

$$A_c = \begin{bmatrix} -a_1 & -a_2 & \cdots & -a_{n-1} & -a_n \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix}, \quad B_c = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad D_u = b_0, \quad D_e = 1$$
$$C_u = [b_1 - a_1 b_0 \quad \cdots \quad b_n - a_n b_0] \quad C_e = [c_1 - a_1 \quad \cdots \quad c_n - a_n]$$

Controllable form: controllability matrix from (A_c, B_c) has full rank, i.e., the system is controllable (input drives the state everywhere in finite time).

Realization: Controllable Form - Example (part 1)

Consider the ARMAX model

$$y_k = \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1} + a_2 z^{-2}} u_k + \frac{1 + c_1 z^{-1}}{1 + a_1 z^{-1} + a_2 z^{-2}} e_k$$

where

$$A(z^{-1}) = 1 + a_1 z^{-1} + a_2 z^{-2},$$

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$$B(z^{-1}) = b_0 + b_1 z^{-1},$$

$$C(z^{-1}) = 1 + c_1 z^{-1}.$$

Using the above formulas, the state space matrix are

$$A_c = \begin{bmatrix} -a_1 & -a_2 \\ 1 & 0 \end{bmatrix}, \quad B_c = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad D_u = b_0, \quad D_e = 1,$$

$$C_u = [b_1 - a_1 b_0 \quad -a_2 b_0], \quad C_e = [c_1 - a_1 \quad -a_2]$$

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$$C_u = [b_1 - a_1 b_0 \quad -a_2 b_0], \quad C_e = [c_1 - a_1 \quad -a_2]$$

corresponding to the transfer function

$$y_k = (C_u(zI - A_c)^{-1} B_c + D_u) u_k + (C_e(zI - A_c)^{-1} B_c + D_e) e_k$$

Realization: Controllable Form - Example (part 2)

Explicit computation lead (for the first term, the second can be similarly checked)

$$\begin{aligned}C_u(zI - A_c)^{-1}B_c + D_u &= [b_1 - a_1b_0 \quad -a_2b_0] \begin{bmatrix} z + a_1 & a_2 \\ -1 & z \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b_0 \\&= [b_1 - a_1b_0 \quad -a_2b_0] \frac{1}{z^2 + a_1z + a_2} \begin{bmatrix} z & -a_2 \\ 1 & z + a_1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b_0 \\&= \frac{(b_1 - a_1b_0)z - a_2b_0 + b_0(z^2 + a_1z + a_2)}{z^2 + a_1z + a_2} \\&= \frac{b_1z + b_0z^2}{z^2 + a_1z + a_2} \\&= \frac{z}{z^2} \frac{b_1 + b_0z}{1 + a_1z^{-1} + a_2z^{-2}} \\&= z^{-1} \frac{b_1 + b_0z}{1 + a_1z^{-1} + a_2z^{-2}} = \frac{b_0 + b_1z^{-1}}{1 + a_1z^{-1} + a_2z^{-2}}\end{aligned}$$

Realization: Observable Canonical Form (Dual)

A dual (observable) state space form with state $x_k \in \mathbb{R}^{n_a}$ is

$$x_{k+1} = A_o x_k + B_u u_k + B_e e_k, \quad y_k = C_o x_k + D_u u_k + D_e e_k.$$

where the **observable canonical matrices** are built as the dual of the controllable companion form, i.e., $A_o = A_c^\top$ and $C_o = B_c^\top$.

$$A_o = A_c^\top = \begin{bmatrix} -a_1 & 1 & 0 & \cdots & 0 \\ -a_2 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ -a_{n-1} & 0 & 0 & \cdots & 1 \\ -a_n & 0 & 0 & \cdots & 0 \end{bmatrix}, \quad C_o = B_c^\top = [1 \quad 0 \quad \cdots \quad 0]$$
$$B_u := \begin{bmatrix} b_1 - a_1 b_0 \\ b_2 - a_2 b_0 \\ \vdots \\ b_n - a_n b_0 \end{bmatrix}, \quad B_e := \begin{bmatrix} c_1 - a_1 \\ c_2 - a_2 \\ \vdots \\ c_n - a_n \end{bmatrix} \quad D_u = b_0, \quad D_e = 1$$

Observable form: observability matrix from (A_o, C_o) has full rank, i.e., the state can be reconstructed from a finite window of outputs.

Realization: Innovation (Kalman) Form

An **innovation-form** (a.k.a. **Kalman form**) state-space model is

$$x_{k+1} = A x_k + B u_k + K e_k, \quad y_k = C x_k + D u_k + e_k.$$

defined by interpreting $e_k = y_k - \hat{y}_{k|k-1}$ as innovation.

For monic ARMAX, a convenient choice, consistent with the controllable realization, is

$$A := A_c, \quad B := B_c, \quad C := (C_u + C_e), \quad D := D_u, \quad K := B_c, \quad D_e := 1,$$

so that

$$\begin{aligned} x_{k+1} &= A_c x_k + B_c u_k + B_c e_k, \\ y_k &= (C_u + C_e) x_k + D_u u_k + e_k. \end{aligned}$$

equivalent to the controllable form.

Innovation form: the noise enters as a *one-step-ahead prediction error*.

Realization: Observer (Predictor) Form

Assuming an innovation-form model

$$x_{k+1} = A x_k + B u_k + K e_k, \quad y_k = C x_k + D u_k + e_k.$$

Eliminate the unmeasured innovation e_k using $e_k = y_k - (C x_k + D u_k)$.

Substitute into the state update to get the **observer / predictor form**:

$$x_{k+1} = (A - KC) x_k + (B - KD) u_k + K y_k, \quad \hat{y}_{k|k-1} = C x_k + D u_k.$$

For consistency with the controllable ARMAX realization:

$$A := A_c, \quad B := B_c, \quad C := (C_u + C_e), \quad D := D_u, \quad K := B_c,$$

Observer (predictor) form: state propagated using the measured output y_k .

Note: this form is not really generative. Rather y_k are assumed to be available.

Realization: Continuous to Discrete (Exact) (part 1)

Consider a general CT model

$$\begin{aligned}\dot{x}(t) &= A x(t) + B u(t) + G w(t) \\ y(t) &= C x(t) + D u(t) + v(t)\end{aligned}$$

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Assume a sampling period $T > 0$ and **zero-order hold (ZOH)** input, the equations for $t \in [kT, (k+1)T]$ leads to

$$x(t) = e^{A(t-kT)} x_k + \int_{kT}^t e^{A(t-\tau)} B u(\tau) d\tau + \int_{kT}^t e^{A(t-\tau)} G w(\tau) d\tau.$$

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and evaluate at $t = (k+1)T$ to

$$x_{k+1} = e^{AT} x_k + \left(\int_0^T e^{A(T-\tau)} B d\tau \right) u_k + \int_0^T e^{A(T-\tau)} G w(kT + \tau) d\tau.$$

Realization: Continuous to Discrete (Exact) (part 2)

With the change of variable $\sigma = T - \tau$ and output sampling, we get:

$$x_{k+1} = A_d x_k + B_d u_k + w_k$$

$$y_k = C x_k + D u_k + v_k.$$

$$A_d := e^{AT}, \quad B_d := \int_0^T e^{A\sigma} B d\sigma, \quad w_k := \int_0^T e^{A\sigma} G w(kT + T - \sigma) d\sigma.$$

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If $w(t)$ is CT white noise with spectral density Q_c , $E[w(t)w(\tau)^\top] = Q_c \delta(t - \tau)$, then w_k is discrete-time white with

$$Q_d := E[w_k w_k^\top] = \int_0^T e^{A\sigma} G Q_c G^\top e^{A^\top \sigma} d\sigma.$$

Note: there are also other (simpler but inexact) discretization techniques, e.g. Forward/Backward Euler etc.

Example of non-stationary process

A process is **stationary** when its statistical properties (mean, variance, autocorrelation) do not change over time. Typically arise from **stable systems**.

- thermal noise in an electrical resistor
- vibrations in mechanical systems
- turbulent flow velocity fluctuations
- noise in sensors

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- turbulent flow velocity fluctuations
- noise in sensors

A process is **non-stationary** when its statistical properties change over time. Often caused by: integration, accumulation, system drift.

- particle under Brownian motion (diffusion process)
- position of a vehicle (tracking navigation)
- financial asset prices

State-Space Derivation of ARIMA(p, d, q) (part 1)

Let $\Delta := 1 - z^{-1}$ (difference operator). An ARIMA(p, d, q) process satisfies

$$A(z^{-1}) \Delta^d y_k = C(z^{-1}) e_k, \quad e_k \sim WN(0, \sigma^2),$$

with monic polynomials

$$A(z^{-1}) = 1 + a_1 z^{-1} + \cdots + a_p z^{-p}, \quad C(z^{-1}) = 1 + c_1 z^{-1} + \cdots + c_q z^{-q}.$$

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$$A(z^{-1}) = 1 + a_1 z^{-1} + \cdots + a_p z^{-p}, \quad C(z^{-1}) = 1 + c_1 z^{-1} + \cdots + c_q z^{-q}.$$

For an ARIMA model, define the stationary differenced series

$$z_k := \Delta^d y_k.$$

Then

$$A(q^{-1}) z_k = C(q^{-1}) e_k$$

is an ARMA(p, q) model for z_k .

State-Space Derivation of ARIMA(p, d, q) (part 2)

Find a n -dim innovation realization ($n = \max\{p, q + 1\}$) for the ARMA for z_k

$$x_{k+1}^z = A_z x_k^z + K_z e_k, \quad z_k = C_z x_k^z + e_k,$$

where (A_z, C_z) are chosen so that the transfer $e \mapsto z$ equals $\frac{C(q^{-1})}{A(q^{-1})}$.

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Add d discrete integrators $s_k^{(1)}, \dots, s_k^{(d)}$ to recover $y_k = s_k^{(d)}$ from $z_k = \Delta^d y_k$

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Equivalently, with $s_k := \begin{bmatrix} s_k^{(1)} & \dots & s_k^{(d)} \end{bmatrix}^\top$ and T_d the “cumulative-sum” matrix:

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Let be $x_k := \begin{bmatrix} (x_k^z)^\top & s_k^\top \end{bmatrix}^\top$, using the ARMA equations

$$x_{k+1} = \begin{bmatrix} A_z & 0 \\ b_d C_z A_z & T_d \end{bmatrix} x_k + \begin{bmatrix} K_z \\ b_d C_z K_z \end{bmatrix} e_k + \begin{bmatrix} 0 \\ b_d \end{bmatrix} e_{k+1},$$
$$y_k = \begin{bmatrix} 0 & e_d^\top \end{bmatrix} x_k.$$

Example: tracking in 2D

Assume random acceleration driving the motion. The particle position motion is

$$\begin{aligned}\mathbf{p}_{k+1} &= \mathbf{p}_k + T\mathbf{v}_k, & \mathbf{p}_k &= \begin{bmatrix} x_k \\ y_k \end{bmatrix} \\ \mathbf{v}_{k+1} &= \mathbf{v}_k + T\mathbf{a}_k, & \mathbf{a}_k &\sim WN(\mathbf{0}, \sigma_a^2 I_2)\end{aligned}$$

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Removing $\mathbf{v}_k$, the position satisfies the ARIMA(0, 2, 0) model

$$\begin{aligned}(1 - z^{-1})^2 \mathbf{p}_k &= \mathbf{p}_k - 2\mathbf{p}_{k-1} + \mathbf{p}_{k-2} = \mathbf{e}_k, & \mathbf{e}_k &\sim WN(\mathbf{0}, Q), & Q &= T^2 \sigma_a^2 I_2 \\ \mathbf{y}_k &= \mathbf{p}_k + \mathbf{w}_k, & \mathbf{w}_k &\sim WN(\mathbf{0}, R) & (\text{measurements}).\end{aligned}$$

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Following the above describe realization procedure, the state space model reads

$$\begin{aligned}\mathbf{x}_{k+1} &= \begin{bmatrix} 1 & 0 & T & 0 \\ 0 & 1 & 0 & T \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \mathbf{x}_k + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{a}_k, & \mathbf{a}_k &\sim WN(\mathbf{0}, \sigma_a^2 I_2). \\ \mathbf{y}_k &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \mathbf{x}_k + \mathbf{w}_k & \mathbf{w}_k &\sim WN(\mathbf{0}, R).\end{aligned}$$

The Kalman Filter: reference state space model

In the following we consider the general state space model of the process $\{\mathbf{y}(t)\}$

$$\mathbf{x}(t+1) = A\mathbf{x}(t) + \mathbf{v}(t)$$

$$\mathbf{y}(t) = C\mathbf{x}(t) + \mathbf{w}(t) \quad \mathbb{E} \begin{bmatrix} \mathbf{v}(t) \\ \mathbf{w}(t) \end{bmatrix} = 0, \quad \text{Var} \begin{bmatrix} \mathbf{v}(t) \\ \mathbf{w}(t) \end{bmatrix} = \begin{bmatrix} Q & S \\ S^T & R \end{bmatrix}, R \succ 0$$

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As *widely* known, the Kalman filter deals with estimation of the (unknown and usually not observable) state process $\{\mathbf{x}(t)\}$ $\implies$ **But why?**

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Note: A, C, Q, S, R could be time varying. $R \succ 0$ for simplicity for now because it helps guaranteeing invertibility of the variance of innovation process $\mathbf{e}(t) = \mathbf{y}(t) - \hat{\mathbf{y}}(t|t-1)$.

The Kalman Filter: why the state $\mathbf{x}(t)$? (part 1)

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1. Prediction of $\{\mathbf{y}(t)\}$ from past measurements.

By observing that $\mathbf{w}(t+1) \perp \{\mathbf{v}(s), \mathbf{w}(s), \mathbf{x}_0, s \leq t\} \supset \mathbf{H}_t^-(\mathbf{y})$

$$\implies \hat{\mathbf{y}}(t+1|t) = \hat{\mathbb{E}}[\mathbf{y}(t+1)|\mathbf{H}_t^-(\mathbf{y})] = C\hat{\mathbb{E}}[\mathbf{x}(t+1)|\mathbf{H}_t^-(\mathbf{y})] = C\hat{\mathbf{x}}(t+1|t)$$

The prediction can be directly build from the predicted state!!

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2. Filtering of $\{\mathbf{z}(t)\}$ from $\{\mathbf{y}(t)\}$

Say we want to filter $\mathbf{z}(t)$ from $\mathbf{y}(t) = \mathbf{z}(t) + \xi(t)$ (ξ “colored noise”).

$$\mathbf{x}(t+1) = \begin{bmatrix} \mathbf{x}_1(t+1) \\ \mathbf{x}_2(t+1) \end{bmatrix} = \begin{bmatrix} A_1 & \\ & A_2 \end{bmatrix} \begin{bmatrix} \mathbf{x}_1(t) \\ \mathbf{x}_2(t) \end{bmatrix} + \begin{bmatrix} \mathbf{v}_1(t) \\ \mathbf{v}_2(t) \end{bmatrix} = A\mathbf{x}(t) + \mathbf{v}(t)$$

$$\mathbf{y}(t) = \mathbf{z}(t) + \xi(t) = \begin{bmatrix} C_1 & C_2 \end{bmatrix} \begin{bmatrix} \mathbf{x}_1(t) \\ \mathbf{x}_2(t) \end{bmatrix} + \begin{bmatrix} I & I \end{bmatrix} \begin{bmatrix} \mathbf{w}_1(t) \\ \mathbf{w}_2(t) \end{bmatrix} = C\mathbf{x}(t) + \mathbf{w}(t)$$

$$\mathbf{z}(t) = \begin{bmatrix} C_1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_1(t) \\ \mathbf{x}_2(t) \end{bmatrix} + \mathbf{w}_1(t) = \bar{C}_1\mathbf{x}(t) + \mathbf{w}_1(t)$$

The Kalman Filter: why the state $\mathbf{x}(t)$? (part 2)

$$\implies \hat{\mathbf{z}}(t|t) = \hat{\mathbb{E}}[\mathbf{z}(t)|\mathbf{H}_t^-(\mathbf{y})] = \overline{\mathbf{C}}_1 \hat{\mathbf{x}}(t|t) + \hat{\mathbf{w}}_1(t|t)$$

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Nonetheless, via double projection (on subspaces)

$$\begin{aligned}\hat{\mathbf{w}}_1(t) &= \hat{\mathbb{E}} \left[\hat{\mathbb{E}} [\mathbf{w}_1(t) | \{\mathbf{x}_0, \mathbf{v}(s-1), \mathbf{w}(s); s \leq t\}] | \mathbf{H}_t^-(\mathbf{y}) \right] \\ &= \hat{\mathbb{E}} [R_1 R^{-1} \mathbf{w}(t) | \mathbf{H}_t^-(\mathbf{y})] \\ &= R_1 R^{-1} \hat{\mathbf{w}}(t|t) \\ &= R_1 R^{-1} (\mathbf{y}(t) - C \hat{\mathbf{x}}(t|t))\end{aligned}$$

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Again, the filtering solution can be directly build for the filtered state!!

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$$\mathbf{x}(t+1) = A\mathbf{x}(t) + \mathbf{v}(t)$$

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Note that $\mathbb{E}[\mathbf{v}(t), \mathbf{w}(t)^T] = S$ but $\mathbf{v}(s) \perp \mathbf{w}(s)$, $s \neq t$.

$$\begin{aligned} \implies \tilde{\mathbf{v}}(t) &= \mathbf{v}(t) - \hat{\mathbb{E}}[\mathbf{v}(t) | \mathbf{H}(\mathbf{w})] = \mathbf{v}(t) - \hat{\mathbb{E}}[\mathbf{v}(t) | \mathbf{w}(t)] \quad \perp \mathbf{w}(t) \\ &= \mathbf{v}(t) - SR^{-1}\mathbf{w}(t) \end{aligned}$$

$$\implies \text{Var} \begin{bmatrix} \tilde{\mathbf{v}}(t) \\ \mathbf{w}(t) \end{bmatrix} = \begin{bmatrix} \tilde{Q} & \\ & R \end{bmatrix}, \quad \tilde{Q} = Q - SR^{-1}S^T$$

$$\implies \mathbf{x}(t+1) = A\mathbf{x}(t) + \mathbf{v}(t) = F\mathbf{x}(t) + SR^{-1}\mathbf{y}(t) + \tilde{\mathbf{v}}(t), \quad F = A - SR^{-1}C$$

with uncorrelated noise $\tilde{\mathbf{v}}(t) \perp \mathbf{w}(t)$.

The Kalman Filter: equations

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① A-priori estimates/predictions (temporal update)

- State: $\hat{\mathbf{x}}(t+1|t) = F\hat{\mathbf{x}}(t|t) + SR^{-1}\mathbf{y}(t)$
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where

- $P(t+1|t), P(t|t)$: the prediction and filtering **errors** variances
- $\Lambda(t) = CP(t|t-1)C^T + R$: innovation ($\mathbf{e}(t) = \mathbf{y}(t) - C\hat{\mathbf{x}}(t|t-1)$) variance
- $L(t) = P(t|t-1)C^T\Lambda(t)^{-1}$: Kalman gain

The Kalman Filter: equations - proof (part 1)

1. **A-priori (prediction) update.** The one-step ahead estimate is obtained by taking the linear conditional expectation given past measurements $\mathbf{H}_t^-(\mathbf{y})$, recalling that $\tilde{\mathbf{v}}(t) \perp \mathbf{w}(t) \implies \tilde{\mathbf{v}}(t) \perp \mathbf{H}_t^-(\mathbf{y})$:

$$\hat{\mathbf{x}}(t+1|t) := \hat{\mathbb{E}}[\mathbf{x}(t+1)|\mathbf{H}_t^-(\mathbf{y})] = F\hat{\mathbb{E}}[\mathbf{x}(t)|\mathbf{H}_t^-(\mathbf{y})] + SR^{-1}\mathbf{y}(t) = F\hat{\mathbf{x}}(t|t) + SR^{-1}\mathbf{y}(t)$$

The corresponding prediction error $\tilde{\mathbf{x}}(t+1|t) = \mathbf{x}(t+1) - \hat{\mathbf{x}}(t+1|t)$ satisfies

$$\tilde{\mathbf{x}}(t+1|t) = F\tilde{\mathbf{x}}(t|t) + \tilde{\mathbf{v}}(t)$$

and, since $\tilde{\mathbf{v}}(t) \perp \tilde{\mathbf{x}}(t|t)$ because

$$\left. \begin{array}{l} \mathbf{x}(t) \in \text{span}\{\mathbf{x}(t_0), \tilde{\mathbf{v}}(s), \mathbf{w}(s); s < t\} \\ \mathbf{H}_t^-(\mathbf{y}) \subset \text{span}\{\mathbf{x}(t_0), \tilde{\mathbf{v}}(s-1), \mathbf{w}(s); s \leq t\} \end{array} \right\} \perp \tilde{\mathbf{v}}(t)$$

its covariance evolves as

$$P(t+1|t) = \text{Var}[\tilde{\mathbf{x}}(t+1|t)] = FP(t|t)F^T + \tilde{Q}$$

The Kalman Filter: equations - proof (part 2)

2. **A-posteriori (measurement) update.** At time $t + 1$ we get $\mathbf{y}(t + 1) = \mathbf{C}\mathbf{x}(t + 1) + \mathbf{w}(t + 1)$. Note that

$$\mathbf{H}_{t+1}^-(\mathbf{y}) = \mathbf{H}_t^-(\mathbf{y}) \oplus \mathbf{H}(\mathbf{e}(t + 1)) \quad (\text{orthogonal sum})$$

being

$$\mathbf{e}(t + 1) = \mathbf{y}(t + 1) - \mathbf{C}\hat{\mathbf{x}}(t + 1|t) = \mathbf{C}\tilde{\mathbf{x}}(t + 1|t) + \mathbf{w}(t + 1)$$

$$\begin{aligned} \implies \hat{\mathbf{x}}(t + 1|t + 1) &= \mathbb{E}[\mathbf{x}(t + 1)|\mathbf{H}_{t+1}^-(\mathbf{y})] = \hat{\mathbf{x}}(t + 1|t) + L(t + 1)\mathbf{e}(t + 1) \\ &\quad \hat{\mathbf{x}}(t + 1|t) + L(t + 1)(\mathbf{y}(t + 1) - \mathbf{C}\hat{\mathbf{x}}(t + 1|t)) \end{aligned}$$

where $L(t + 1) = \text{Cov}[\mathbf{x}(t + 1), \mathbf{e}(t + 1)]\text{Var}[\mathbf{e}(t + 1)]^{-1}$ with

$$\begin{aligned} \text{Cov}[\mathbf{x}(t + 1), \mathbf{e}(t + 1)] &= \text{Cov}[\mathbf{x}(t + 1), \tilde{\mathbf{x}}(t + 1|t)]\mathbf{C}^T + \text{Cov}[\mathbf{x}(t + 1), \mathbf{w}(t + 1)] \\ &= \text{Cov}[\tilde{\mathbf{x}}(t + 1|t), \tilde{\mathbf{x}}(t + 1|t)] = \mathbf{P}(t + 1|t) \end{aligned}$$

$$\text{Var}[\mathbf{e}(t + 1)] = \mathbf{\Lambda}(t + 1) = \mathbf{C}\mathbf{P}(t + 1|t)\mathbf{C}^T + \mathbf{R}, \quad \text{since } \tilde{\mathbf{x}}(t + 1|t) \perp \mathbf{w}(t + 1)$$

The Kalman Filter: equations - proof (part 3)

The covariance update follows by noting that

$$\begin{aligned}\mathbf{x}(t+1) - \hat{\mathbf{x}}(t+1|t+1) &= \mathbf{x}(t+1) - \hat{\mathbf{x}}(t+1|t) - L(t+1)\mathbf{e}(t+1) \\ \implies \tilde{\mathbf{x}}(t+1|t+1) &= \tilde{\mathbf{x}}(t+1|t) - L(t+1)\mathbf{e}(t+1) \\ \implies \tilde{\mathbf{x}}(t+1|t+1) + L(t+1)\mathbf{e}(t+1) &= \tilde{\mathbf{x}}(t+1|t)\end{aligned}$$

and being $\tilde{\mathbf{x}}(t+1|t+1) \perp \mathbf{e}(t+1)$

$$\begin{aligned}P(t+1|t+1) + L(t+1)\Lambda(t+1)L(t+1)^T &= P(t+1|t) \\ \implies P(t+1|t+1) &= P(t+1|t) - L(t+1)\Lambda(t+1)L(t+1)^T \\ &= P(t+1|t) - P(t+1|t)C^T\Lambda(t+1)^{-1}CP(t+1|t)\end{aligned}$$

Considerations on error covariance update

- 1 To avoid/reduce numerical ill-conditioning it is sometimes preferred to express the covariance update in different forms

$$\begin{aligned} P(t+1|t+1) &= P(t+1|t) - P(t+1|t)C^T\Lambda(t+1)^{-1}CP(t+1|t) \\ &= (I - L(t+1)C)P(t+1|t) \end{aligned}$$

$$\begin{aligned} \text{symmetrized form} &= [I - L(t+1)C]P(t+1|t)[I - L(t+1)C]^T + \\ &\quad L(t+1)RL(t+1)^T \end{aligned}$$

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- 2 Note that **ALL** the **covariance computations** (error and innovation) as well as the **Kalman Gain** do not explicitly depend on the measurements and can be pre-computed **offline**. This is the major computational burden!
- 3 Conversely, the **state update and prediction** depend on the measurements and must be computed **online**.

The Kalman Predictor

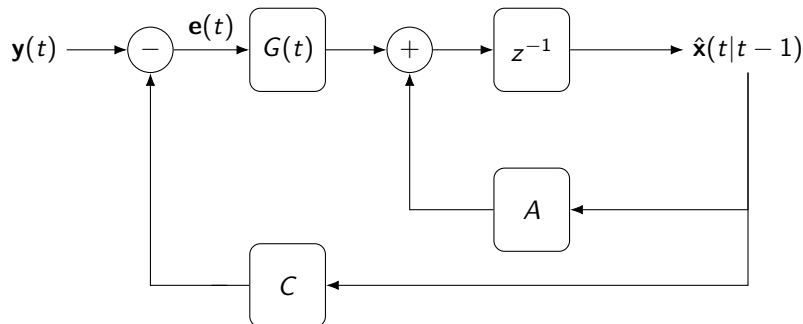
It is often useful to express the dynamics of the predictor which allows to express the next state prediction as function of the previous without computing the filtered value. That is (after some manipulation)

$$\begin{aligned}\hat{\mathbf{x}}(t+1|t) &= F(I - L(t)C)\hat{\mathbf{x}}(t|t-1) + (FL(t) + SR^{-1})\mathbf{y}(t) \\ &= (F - K(t)C)\hat{\mathbf{x}}(t|t-1) + (K(t) + SR^{-1})\mathbf{y}(t) \\ &= \Gamma(t)\hat{\mathbf{x}}(t|t-1) + G(t)\mathbf{y}(t) \\ &= A\hat{\mathbf{x}}(t|t-1) + G(t)[\mathbf{y}(t) - C\hat{\mathbf{x}}(t|t-1)] \\ &= A\hat{\mathbf{x}}(t|t-1) + G(t)\mathbf{e}(t)\end{aligned}$$

$$\begin{aligned}P(t+1|t) &= F[P(t|t-1) - P(t|t-1)C^T\Lambda(t)^{-1}CP(t|t-1)]F^T + \tilde{Q} \\ &= F[I - L(t)C]P(t|t-1)[I - L(t)C]^T F^T + FL(t)RL^T(t)F^T + \tilde{Q}\end{aligned}$$

where $K(t) := FL(t)$, $\Gamma(t) := F - K(t)C$, $G(t) := K(t) + SR^{-1}$.

Equivalent closed loop interpretation



where the closed-loop dynamics is described by

$$\Gamma(t) = A - G(t)C = F - K(t)C$$

Hence its properties characterize the filter/predictor dynamics.

Filtered Estimates

Similarly (for $S = 0$ for simplicity)

$$\begin{aligned}\hat{\mathbf{x}}(t+1|t+1) &= A\hat{\mathbf{x}}(t|t) + L(t+1)[\mathbf{y}(t+1) - CA\hat{\mathbf{x}}(t|t)] \\ P(t+1|t+1) &= [I - L(t+1)C]P(t+1|t)\end{aligned}$$

In this case the closed-loop dynamical behavior is described by

$$\Gamma_{\text{flt}}(t) = [I - L(t+1)C]A$$

where before, with $S = 0$, was $\Gamma_{\text{pred}}(t) = A[I - L(t)C]$ which suggest an almost identical asymptotic dynamic behavior.

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Note that prediction and filtered estimates can directly be computed from the original equations. It is sometimes convenient to get the explicit expressions in order to better understand the role of the measurements in the update.

Kalman Filter with Input (part 1)

Reference model (with usual assumptions on noise and initial state):

$$\begin{aligned}\mathbf{x}(t+1) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) + \mathbf{v}(t) = \mathbf{F}\mathbf{x}(t) + \mathbf{S}\mathbf{R}^{-1}\mathbf{y}(t) + \mathbf{B}\mathbf{u}(t) + \tilde{\mathbf{v}}(t) \\ \mathbf{y}(t) &= \mathbf{C}\mathbf{x}(t) + \mathbf{w}(t) + \mathbf{D}\mathbf{u}(t)\end{aligned}$$

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Easy: if $\mathbf{u}(t)$ **known and deterministic**

$$\begin{aligned}\hat{\mathbf{x}}(t+1|t) &= F\hat{\mathbf{x}}(t|t) + SR^{-1}\mathbf{y}(t) + B\mathbf{u}(t) \\ P(t+1|t) &= \dots \textit{unchanged} \\ \mathbf{e}(t) &= y(t) - C\hat{\mathbf{x}}(t|t-1) - D\mathbf{u}(t)\end{aligned}$$

$\Rightarrow$ the deterministic input does not affect the uncertainty, only the trajectory.

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Harder: *but what if $\mathbf{u}(t)$ is a process?* E.g. a (linear) causal function (H) of $\mathbf{H}_t^-(\mathbf{y})$ plus a process uncorrelated from the entire noise ($\mathbf{v}, \mathbf{w}$) history, say

$$\mathbf{u}(t) = H(\mathbf{y}^t) + \mathbf{r}(t)$$

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In this case, even if $H \equiv 0$, it is in general not possible to assume $\mathbf{u}(t) \perp \mathbf{x}_0$. Even if $\mathbf{u}(t)$ is observable, $\hat{\mathbf{x}}$ based on joint observations $(\mathbf{y}, \mathbf{u})$ must be obtained by considering the joint state model thus deriving augmented KF equations.

Kalman Filter with Input (part 2)

Assume the plant and input are described by

$$\begin{cases} \mathbf{x}_1(t+1) &= A\mathbf{x}_1(t) + B\mathbf{u}(t) + \mathbf{v}_1(t) \\ \mathbf{y}(t) &= C\mathbf{x}(t) + \mathbf{w}_1(t) \end{cases}$$
$$\begin{cases} \mathbf{x}_2(t+1) &= F\mathbf{x}_2(t) + G\mathbf{y}(t) + \mathbf{v}_2(t) \\ \mathbf{u}(t) &= H\mathbf{x}_2(t) + J\mathbf{y}(t) + \mathbf{w}_2(t) \end{cases}$$

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This gives rise to the augmented state model to which standard KF can be applied

$$\begin{aligned} \mathbf{x}(t+1) &= \underbrace{\begin{bmatrix} A + BJC & BH \\ GC & F \end{bmatrix}}_A \mathbf{x}(t) + \underbrace{\begin{bmatrix} \mathbf{v}_1(t) \\ \mathbf{v}_2(t) \end{bmatrix} + \begin{bmatrix} BJ & B \\ G & \end{bmatrix} \begin{bmatrix} \mathbf{w}_1(t) \\ \mathbf{w}_2(t) \end{bmatrix}}_{\bar{\mathbf{v}}(t)} \\ \begin{bmatrix} \mathbf{y}(t) \\ \mathbf{u}(t) \end{bmatrix} &= \underbrace{\begin{bmatrix} C \\ JC & H \end{bmatrix}}_C \mathbf{x}(t) + \underbrace{\begin{bmatrix} I \\ J & I \end{bmatrix}}_D \underbrace{\begin{bmatrix} \mathbf{w}_1(t) \\ \mathbf{w}_2(t) \end{bmatrix}}_{\mathbf{w}(t)} \end{aligned}$$

The Kalman Filter: information form

1 Information variables

- Information matrix: $Y(t|t) \triangleq P(t|t)^{-1}$
- Information state: $\xi(t|t) \triangleq Y(t|t)\hat{x}(t|t)$

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- State: $\xi(t+1|t) = Y(t+1|t) (F Y(t|t)^{-1} \xi(t|t) + SR^{-1} \mathbf{y}(t))$
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where

- $Y(t|t-1), Y(t|t)$: the prediction and filtering **information** matrices
- $\xi(t|t-1), \xi(t|t)$: the corresponding information states
- $\hat{\mathbf{x}}(t|t) = Y(t|t)^{-1} \xi(t|t)$

Kalman filter vs. Information filter

Standard Kalman filter

- Propagates

$$\hat{\mathbf{x}}(t|t), \quad P(t|t)$$

- Measurement update through

$$\Lambda(t) = CP(t|t-1)C^T + R$$

- Gain computation required:

$$L(t) = P(t|t-1)C^T\Lambda(t)^{-1}$$

- Natural for moderate state dimension and direct state reconstruction at each t
- Direct Covariance update

Key relation

$$Y(t|t) = P(t|t)^{-1}, \quad \xi(t|t) = Y(t|t)\hat{\mathbf{x}}(t|t)$$

Information filter

- Propagates

$$Y(t|t), \quad \xi(t|t)$$

- Measurement update is additive:

$$Y^+ = Y^- + C^T R^{-1} C$$

$$\xi^+ = \xi^- + C^T R^{-1} y$$

- No explicit Kalman gain
- Natural for multi-sensor fusion and sparse large-scale problems
- Particularly convenient when measurements are many and independent

Why information form can be advantageous

Example: fusion of many independent sensors

Assume N sensors observe the same state:

$$y_i(t) = C_i x(t) + v_i(t), \quad v_i(t) \sim \mathcal{N}(0, R_i), \quad i = 1, \dots, N$$

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Main advantage

Updates are simple accumulation of information: more scalable and simpler.

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In the KF context, we deal with estimation of a time-varying random vector based on data and prior information. Hence the relevant bound is what is usually referred to as the **posterior (Bayesian) CRLB**

$$\mathcal{I}(t|t) = -\mathbb{E}[\nabla_{\mathbf{x}}^2 \log p(\mathbf{x}(t) | \mathbf{H}_t^-(\mathbf{y})) \mid \mathbf{H}_t^-(\mathbf{y})] \quad \text{Fisher Information Matrix}$$

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- For any unbiased estimator $\hat{\mathbf{x}}_u(t|t)$ of $\mathbf{x}(t)$ from $\mathbf{H}_t^-(\mathbf{y})$:

$$\mathbb{E}[\hat{\mathbf{x}}_u(t|t) - \mathbf{x}(t)] = 0, \quad \text{Cov}[\hat{\mathbf{x}}_u(t|t) - \mathbf{x}(t)] \succeq \mathcal{I}(t|t)^{-1}$$

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- Under the linear-Gaussian assumptions used for KF:

$$\mathcal{I}(t|t) = \mathcal{Y}(t|t) = P(t|t)^{-1} \implies \text{Cov}[\hat{\mathbf{x}}_u(t|t) - \mathbf{x}(t)] \succcurlyeq P(t|t)$$

and the Kalman filter attains the bound:

$$\text{Cov}[\tilde{\mathbf{x}}(t|t)] = \text{Cov}[\mathbf{x}(t) - \hat{\mathbf{x}}(t|t)] = P(t|t).$$

Cramer-Rao limit (KF): proof of bound attainability

Under the linear-Gaussian assumptions $\mathbf{x}(t) | \mathbf{H}_t^-(\mathbf{y}) \sim \mathcal{N}(\hat{\mathbf{x}}(t|t), P(t|t))$, where

$$\hat{\mathbf{x}}(t|t) = \hat{\mathbb{E}}[\mathbf{x}(t) | \mathbf{H}_t^-(\mathbf{y})] \equiv \mathbb{E}[\mathbf{x}(t) | \mathbf{H}_t^-(\mathbf{y})] \quad (\text{the KF})$$

For any estimator $\hat{\mathbf{x}}_u(t|t)$ based on $\mathbf{H}_t^-(\mathbf{y})$, define

$$d(t) := \hat{\mathbf{x}}_u(t|t) - \hat{\mathbf{x}}(t|t), \quad \tilde{\mathbf{x}}(t|t) := \mathbf{x}(t) - \hat{\mathbf{x}}(t|t).$$

By the orthogonality principle of conditional expectation, $\mathbb{E}[\tilde{\mathbf{x}}(t|t) d(t)^T] = 0$.

Hence

$$\begin{aligned} \text{Cov}[\mathbf{x}(t) - \hat{\mathbf{x}}_u(t|t)] &= \text{Cov}[\tilde{\mathbf{x}}(t|t) - d(t)] \\ &= \text{Cov}[\tilde{\mathbf{x}}(t|t)] + \text{Cov}[d(t)] \\ &= P(t|t) + \text{Cov}[d(t)] \succeq P(t|t). \end{aligned}$$

Equality holds iff $\text{Cov}[d(t)] = 0$ (i.e., $d(t) = 0$ a.s.), namely $\hat{\mathbf{x}}_u(t|t) = \hat{\mathbf{x}}(t|t)$

Asymptotic Behavior

TODO: add asymptotic behavior

9 Nonlinear Theory

Linear vs Nonlinear Kalman Filtering

- **Linear Kalman Filter (KF):** Assumes linear system dynamics and measurement models.

Linear vs Nonlinear Kalman Filtering

- **Linear Kalman Filter (KF):** Assumes linear system dynamics and measurement models.
- **Nonlinear versions:**
 - **Extended Kalman Filter (EKF):** Linearizes nonlinear models using Taylor series expansion.
 - **Unscented Kalman Filter (UKF):** Uses deterministic sampling to capture mean and covariance of nonlinear transformations.
 - **Other variants:** E.g. Particle Filters (PFs), use random sampling to approximate any underlying distribution.

Linear vs Nonlinear Kalman Filtering

- **Linear Kalman Filter (KF):** Assumes linear system dynamics and measurement models.
- **Nonlinear versions:**
 - **Extended Kalman Filter (EKF):** Linearizes nonlinear models using Taylor series expansion.
 - **Unscented Kalman Filter (UKF):** Uses deterministic sampling to capture mean and covariance of nonlinear transformations.
 - **Other variants:** E.g. Particle Filters (PFs), use random sampling to approximate any underlying distribution.

Estimator	Model	Assumed Distribution	Computational Cost
KF	Linear	Gaussian	Low
EKF	Locally Linear	Gaussian	Low/Medium (depends on Jacobians)
UKF	Nonlinear	Gaussian	Medium
PFs	Nonlinear	Arbitrary	High

Extended Kalman Filter (EKF): essentials and notation

- Linearizes nonlinear functions $f(x, u)$ and $h(x)$ around current estimate.
- Same equations as linear KF, but with linearized matrices.
- Limitations: Linearization errors, Jacobian computation.

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Consider a discrete-time⁶ non-linear state model of the form

$$\begin{aligned}\mathbf{x}(k+1) &= f(\mathbf{x}(k), \mathbf{u}(k)) + \mathbf{v}(k), \quad \mathbf{v} \sim \mathcal{N}(0, Q) \\ \mathbf{y}(k) &= h(\mathbf{x}(k)) + \mathbf{w}(k), \quad \mathbf{w} \sim \mathcal{N}(0, R)\end{aligned}$$

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under the usual assumptions $\mathbf{v} \perp \mathbf{w} \perp \mathbf{x}_0$, and denote the Jacobian matrices

$$F_k = \left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_k|k} \quad H_k = \left. \frac{\partial h}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_{k+1}|k}$$

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EKF: discrete-time equations

1 Prior estimates

- State: $\hat{\mathbf{x}}(k+1|k) = f(\hat{\mathbf{x}}(k|k), \mathbf{u}(k))$
- Variance: $P(k+1|k) = F_k P(k|k) F_k^T + Q,$

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- State: $\hat{\mathbf{x}}(k+1|k+1) = \hat{\mathbf{x}}(k+1|k) + L(k+1) [\mathbf{y}(k+1) - h(\hat{\mathbf{x}}(k+1|k))]$
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Nonlinearity used to properly propagate the prior estimate and the measurement model to compute the pseudo innovation.

EKF: generalization - continuous-discrete equations

The EKF is also used in more general setups of continuous-discrete systems.

⁷Matrix satisfying the Volterra equation which, for constant F , equals $\Phi(t,s) = e^{F(t-s)}$.

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Consider a continuous-time state-model with sampled measurements at discrete instants t_k (possibly non uniform)

$$\begin{aligned}\dot{\mathbf{x}}(t) &= f(\mathbf{x}(t)) + \xi(t) \\ \mathbf{y}(t_k) &= h(\mathbf{x}(t_k)) + \mathbf{w}(t_k)\end{aligned}$$

being ξ white noise of intensity $Q \geq 0$; $\mathbf{w}(t_k)$ white of variance $R > 0$ and $\{\xi(t)\} \perp \{\mathbf{w}(t_k)\} \perp \mathbf{x}(t_0)$. $f, h \in \mathbb{C}^1$.

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Let $\Phi(t, s, \bar{\mathbf{x}})$ be the fundamental matrix⁷ associated to the time-varying linear system $\dot{\mathbf{z}} = F(\bar{\mathbf{x}}(t))\mathbf{z}$ where $F(\bar{\mathbf{x}}(t)) = \frac{\partial f}{\partial \mathbf{x}}|_{\mathbf{x}=\bar{\mathbf{x}}(t)}$ and define

$$Q(t_k) := \int_{t_k}^{t_{k+1}} \Phi(t_{k+1}, s, \bar{\mathbf{x}}) Q \Phi(t_{k+1}, s, \bar{\mathbf{x}})^T ds$$

Under these conditions, the equations generalize as follows.

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EKF: generalization - continuous-discrete equations

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- State:

- Resolve $\dot{\bar{\mathbf{x}}}(t) = f(\bar{\mathbf{x}}(t))$, $t \in [t_k, t_{k+1}]$, $\bar{\mathbf{x}}(t_k) = \hat{\mathbf{x}}(k|k)$
- Set $\hat{\mathbf{x}}(k+1|k) = \bar{\mathbf{x}}(t_{k+1})$

- Variance:

- $P(k+1|k) = \Phi(k|k)P(k|k)\Phi(k|k)^T + Q(k)$, $\Phi(k|k) := \Phi(k+1, k, \hat{\mathbf{x}}(k|k))$
- Alternative CT Riccati: $\dot{P}(t) = F(t)P(t) + P(t)F(t)^T + Q(t)$

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- $H(k+1|k) := H(\hat{\mathbf{x}}(k+1|k)) = \frac{\partial h}{\partial \mathbf{x}}|_{\mathbf{x}=\hat{\mathbf{x}}(k+1|k)}$

• Variance: $P(k+1|k+1) = (I - L(k+1)H(k+1|k))P(k+1|k)$

with initial conditions $\hat{\mathbf{x}}(0, -1) = \mathbb{E}[\mathbf{x}(t_0)]$, $P(0|-1) = \text{Var}[\mathbf{x}(t_0)]$

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To avoid explicit integration, when the signals are sufficiently slow w.r.t. the sampling time, a full DT scheme is used, by discretizing the CT dynamics, e.g.,

$$\mathbf{x}(k+1) = \hat{f}_k(\mathbf{x}(k)) \stackrel{\text{Euler}}{=} \mathbf{x}(k) + (t_{k+1} - t_k)f(\mathbf{x}(k))$$

also used to propagate the prior estimated state in the filter equations

$$\hat{\mathbf{x}}(k+1|k) = \hat{f}_k(\hat{\mathbf{x}}(k|k))$$

All other equations remain the same assuming now $\Phi(k|k) = \frac{\partial \hat{f}_k}{\partial \mathbf{x}}|_{\mathbf{x}=\hat{\mathbf{x}}(k|k)}$.

Unscented Kalman Filter (UKF)

- Uses sigma points to capture higher-order moments: for any random vector $\mathbf{x}$, sigma points are a set of any N vectors satisfying:
 - first order: weighted average of points corresponds to $\mathbb{E}[\mathbf{x}]$
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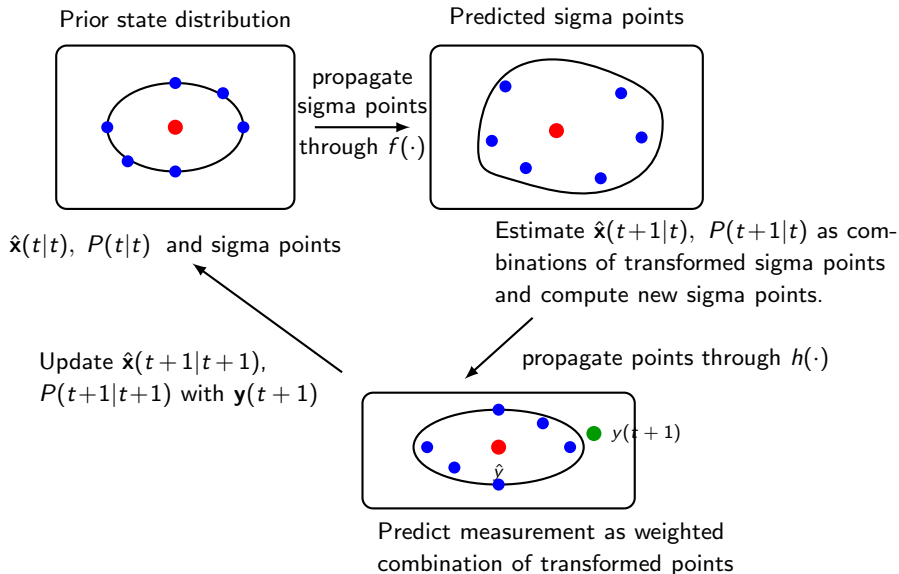
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UKF equations are not difficult but heavy in notation. We'll see them in code rather than on slides.

UKF: graphical representation



Particle Filters (PFs)

Particle Filters are a family rather than a single estimation algorithm.

General idea:

- use many samples/particle
- use random sampling to approximate any underlying distribution
- often exploit a *genetic* approach consisting of a selection/updating step alternate with a mutation/prediction step

Common implementations are:

- Markov Chain Monte Carlo
- Sequential Importance Sampling (SIS)
- Sequential Importance Resampling (SIR)

3 Adds-on

A Practical Recipe for Tuning a Kalman Filter (part 1)

Goal: essentially, tune the two covariances that matter most:

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- Hold the system still or run in a regime with known truth
- Compute sample covariance of sensor error

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3. Check the innovations

$$\nu_k = y_k - H\hat{x}_k^-, \quad S_k = HP_k^- H^\top + R$$

A good filter has innovations that are approximately white, zero-mean, and with the expected size (covariance close to S_k).

A Practical Recipe for Tuning a Kalman Filter (part 2)

4. Tune using observed behavior

Symptom	Adjustment
Estimate too noisy	increase R or decrease Q
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Bias / drift not tracked	increase Q on bias states

Best practical rule: fix R from measurements, then tune only Q until the filter is as fast as possible without visibly fitting noise.

Example sweep

```
for alpha in [1e-3, 1e-2, ..., 1e3]:  
    Q = alpha * Q0  
    run filter  
    inspect innovation variance + tracking lag
```

Fixed-Interval Rauch-Tung-Striebel Smoother

Objective of smoothing:

- Fixed-lag: provide an almost real-time (slightly delayed) improved estimate at $k - N$ using a sliding window of measurements up to k .
- Fixed-interval: provide the best estimate using all available measurements up to a certain instant n .

⁸There are also other smoother implementation e.g., the Modified Bryson-Frazier, which we are not going to cover.

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- forward pass: exactly equal to the standard KF equations to compute $\hat{\mathbf{x}}(k|k-1)$, $\hat{\mathbf{x}}(k|)$, $P(k|k-1)$ and $P(k|k)$.

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 - $\hat{\mathbf{x}}(k|n) = \hat{\mathbf{x}}(k|k) + C_k (\hat{\mathbf{x}}(k+1|n) - \hat{\mathbf{x}}(k+1|k))$
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Note that, conversely to fixed-lag smoothers, fixed-interval are **offline** algorithms as they require all future measurements up to n .

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LQG control and its connection to KF

Problem setup (stochastic optimal control)

$$x_{t+1} = Ax_t + Bu_t + w_t, \quad y_t = Cx_t + v_t, \quad w_t \sim \mathcal{N}(0, Q), \quad v_t \sim \mathcal{N}(0, R)$$

Objective (LQG cost): $J = \mathbb{E} \left[\sum_{t=0}^T x_t^T Q_x x_t + u_t^T R_u u_t \right]$

Key result (Separation principle): Optimal control = LQR + KF

- **State estimation (Kalman filter):** $\hat{x}(t|t) = \mathbb{E}[x_t | \mathbf{H}_t^-(y)]$
- **Control law (LQR):** $u_t = -K_t \hat{x}(t|t)$ (Controller uses $\hat{x}$, not x)

Dual Riccati equations (forward + backward)

- KF covariance: $P_{t+1} = AP_t A^T + Q - AP_t C^T (CP_t C^T + R)^{-1} CP_t A^T$
- LQR cost2go: $S_t = A^T S_{t+1} A + Q_x - A^T S_{t+1} B (B^T S_{t+1} B + R_u)^{-1} B^T S_{t+1} A$

KF (estimation)	LQR (control)
A, C	A^T, B^T
Q, R	Q_x, R_u
P_t	S_t
K_t^{KF}	K_t^{LQR}

Interpretation

- KF: optimal *information extraction*
- LQR: optimal *control action*
- LQG: optimal closed-loop system

Additional resources

- <https://github.com/mintisan/awesome-kalman-filter?tab=readme-ov-file>
- <https://github.com/rlabbe/Kalman-and-Bayesian-Filters-in-Python>
- <https://kalmanfilter.net/>

End! Thank you!